

Software Requirement Specifications

<LIBRARY MANAGEMENT SYSTEM>

Project Code: Group 2_lms

Internal Advisor:Dr. Muhammad Ilyas

External Advisor:Dr. Muhammad Ilyas

Project Manager:Dr. Muhammad Ilyas

Project Team:Shahzeen Fatima, Zenab Noor, Saba Maryam

Submission Date:Sept. 15, 2025

Project Manager's Signature

Document Information

Category	Information
Customer	University of Sargodha
Project	Library management system
Document	Requirement Specifications
Document Version	1.0
Identifier	PGBH01-2003-RS
Status	Draft
Author(s)	Shahzeen fatima,Zenab noor ,Saba maryam
Approver(s)	Dr. Muhammad Ilyas
Issue Date	Sept. 15, 2025
Document Location	
Distribution	1. Advisor 2. Dr. Muhammad Ilyas

Definition of Terms, Acronyms and Abbreviations

Term	Description
ASP	Active Server Pages
RS	Requirements Specifications
LMS	Library Management System
IT	Information Technology
LAN	Local Area Network
DBMS	Database Management System
OS	Operating System
MY Sql	My Structured Query Language

Table of Contents

1. INTRODUCTION	4
1.1 <i>Purpose of Document</i>	4
1.2 <i>Project Overview</i>	4
1.3 <i>Scope</i>	4
2. OVERALL SYSTEM DESCRIPTION	4
2.1 <i>User characteristics</i>	4
2.2 <i>Operating environment</i>	5
2.3 <i>System constraints</i>	5
3. EXTERNAL INTERFACE REQUIREMENTS	5
3.1 <i>Hardware Interfaces</i>	5
3.2 <i>Software Interfaces</i>	5
3.3 <i>Communications Interfaces</i>	6
4. FUNCTIONAL REQUIREMENTS	6
5. NON-FUNCTIONAL REQUIREMENTS	7
5.1 <i>Performance Requirements</i>	7
5.2 <i>Safety Requirements</i>	8
5.3 <i>Security Requirements</i>	8
5.4 <i>User Documentation</i>	8
6. ASSUMPTIONS AND DEPENDENCIES	8
7. REFERENCES	8
8. APPENDICES	9

1. Introduction

1.1 Purpose of Document

The purpose of this SRS is to clearly describe all requirements of the Library Management System. It explains what the system should do, how it will function, and the features it is expected to provide. This document serves as a guide for the project team so that developers, advisors, and users share the same understanding. It ensures that the final software matches user needs and works as expected. The intended audience includes the project team, developers, internal/external advisors, and library management personnel.

1.2 Project Overview

The Library Management System is designed to make everyday library tasks easier and more organized. Instead of relying on handwritten registers, the system keeps all information like books, users, and borrowing records stored digitally. This helps librarians quickly add new books, update details, issue books, and manage returns without any confusion. It keeps digital records of books, students, teachers, and borrowing history, reducing the need for paper-based work and lowering the chances of errors.

1.3 Scope

The system will help librarians to manage books and user information in an organized way.

What the system will do:

- Add, update, and delete book records.
- Manage user details for students, teachers, and admins.
- Issue books and record returns with dynamic duration and volume limits.
- Track overdue books, automatically compute fines, and send advance reminders.
- Allow book reservation for future issuance.
- Generate basic reports of library activity.

What the system will not do:

- It will not handle financial payment transactions for fines directly (only tracking/computation).
- It will not manage external inter-library loans outside the local university campus ecosystem.

2. Overall System Description

The Library Management System will be a computer-based application used in schools, colleges, or universities to manage daily library tasks. It will run in a controlled environment such as a library or computer lab and will be operated mainly by librarians to manage books and user information.

2.1 User characteristics

- **Librarian / Admin (Primary User):** Responsible for adding, updating, and deleting book records. Manages user accounts for students and teachers. Issues and receives books, tracks overdue items, and holds full access to all system features. This is the most critical user class.

- **Student / Teacher (Secondary Users):** Can view available books, check borrowing status, request books for borrowing, and reserve books. They have limited access and cannot modify base library or user records.

2.2 Operating environment

- **Hardware Platform:** Standard PC or laptop equipped with a keyboard, mouse, and monitor.
- **Operating Systems:** Windows or Linux operating systems.
- **Coexisting Software Components:** MySQL Database Server, standard web browsers.
- **Network Infrastructure:** Operates locally on a Local Area Network (LAN); internet connectivity is optional.

2.3 System constraints

- **Software constraints:** Must remain fully compatible with Windows/Linux and MySQL.
- **Hardware constraints:** Must execute efficiently on normal, standard-specification library PCs.
- **Legal constraints:** User personal data and student records must be securely protected from leaks.
- **Cultural constraints:** The interface language should be simple, intuitive, and easy to comprehend.
- **User constraints:** Explicit role boundaries must be enforced via multi-level authentication accounts.

3. External Interface Requirements

This section is intended to specify any requirements that ensure that the new system will connect properly to external components. Place a context diagram showing the external interfaces at a high level of abstraction.

3.1 Hardware Interfaces

Works with standard PC/laptop hardware architectures. Employs basic input devices (keyboard and mouse) for data entries. Utilizes standard display monitors to output visual interfaces and text metadata.

3.2 Software Interfaces

- **Database Management System:** Connects directly with a MySQL database instance.
- **Operating System Environment:** Interacts with Windows or Linux OS platforms.

- **Client Interface:** Renders UI screens through modern web browsers or an independent app wrapper.

3.3 Communications Interfaces

Uses standard Local Area Network (LAN) infrastructure communication protocols. Features an optional web stack to support digital notification deliveries (automated alerts). Employs reliable synchronization mechanisms to prevent concurrent write collisions.

4. Functional Requirements

Functional requirements includes

4.1 Book Management

- 1) The system shall allow librarians to add, update, search, and remove book records.
- 2) Each book shall have a unique identification number.
- 3) The system shall store complete book details, including title, author(s), ISBN, publication date, publisher, version, and serial number.
- 4) The system shall maintain and track the number of available copies of each book.

4.2 User Management

- 1) The system shall maintain separate profiles for students and teachers/admins.
- 2) The system shall allow authorized personnel to create, update, and manage user accounts.
- 3) Student records shall include personal and academic information such as name, DOB, class, session, contact details, and address.
- 4) Teacher and admin accounts shall include credentials and access permissions based on their roles.

4.3 Book Issuing

- 1) The system shall allow books to be issued to students, teachers, and administrators.
- 2) The system shall ensure that a specific book copy can only be issued to one user at a time.
- 3) The system shall enforce borrowing limits based on user type.
- 4) The system shall automatically record issue dates and due dates.

4.4 Book Return Processing

- 1) The system shall allow users to return borrowed books.
- 2) The system shall update book availability immediately after a return is processed.
- 3) The system shall maintain a history of issued and returned books.

4.5 Borrowing Rules Enforcement

- 1) The system shall restrict students to a maximum of 5 active book loans.
- 2) The system shall restrict teachers and administrators to a maximum of 8 active book loans.
- 3) The system shall assign a borrowing period of 10 days for students and 20 days for teachers/admins.

4.6 Reservation Management

- 1) The system shall allow users to reserve books that are currently unavailable.
- 2) The system shall maintain a reservation queue and allocate books according to reservation priority.
- 3) Reserved books shall not be available for general checkout until the reservation is fulfilled or cancelled.

4.7 Reminder and Notification System

- 1) The system shall automatically send reminders to users two days before the due date of a borrowed book.
- 2) The system shall notify users about reservation availability and overdue books.

4.8 Fine Management

- 1) The system shall automatically calculate overdue fines at the rate of 5 rupees per day.
- 2) The system shall maintain a record of outstanding and paid fines for each user.
- 3) The system shall generate fine reports when required.

4.9 Search and Inquiry Services

- 1) The system shall provide search functionality for books using title, author, ISBN, publisher, or other details.
- 2) Users shall be able to view book availability and reservation status.

4.10 Reporting and Record Keeping

- 1) The system shall generate reports related to issued books, returned books, overdue items, reservations, and fines.
- 2) The system shall maintain transaction records for auditing and tracking purposes.

4.11 Authentication and Access Control

- 1) The system shall require users to log in using authorized credentials.
- 2) The system shall provide different access levels for students, teachers, librarians, and administrators.
- 3) The system shall protect sensitive user and library data from unauthorized access.

5. Non-functional Requirements

5.1 Performance Requirements

- **Response Latency:** The system interface must respond quickly and smoothly to lookups.
- **Concurrency:** The architecture must cleanly handle multi-user interactions simultaneously.
- **Data Accuracy:** Calendar calculations and borrowing balances must be error-free.

5.2 Safety Requirements

- **Data Persistence:** Implement automated local backup systems to prevent critical data loss.
- **Stability:** Maintain software stability, minimizing runtime defects and preventing sudden crashes.

5.3 Security Requirements

Restrict admin panels strictly to authorized librarians.Route students and teachers through non-destructive data viewing layers. Encrypt passwords and hide member contact details from unauthenticated entities.

5.4 User Documentation

- **User Manual:** Explicit procedural layout tailored for onboarding administrative librarians.
- **Online Help:** Micro-captions and hover tooltips injected directly within core menu buttons.
- **Tutorials:** Step-by-step visual training modules detailing file creation and asset allocation entries.

6. Assumptions and Dependencies

- **Operator Skill Level:** Staff operators hold necessary basic desktop navigation and IT competency.
- **Infrastructure Availability:** On-premise electricity setups and local area network configurations are operational.
- **Software Interoperability:** OS layers maintain complete baseline driver alignment with MySQL database assets.
- **System Maintenance:** Periodic execution of technical data cleanups to keep operational lag minimal.

7. References

This section should provide a complete list of all documents referenced at specific point in time. Each document should be identified by title, report number (if applicable), date, and publishing organization. Specify the sources from which the references can be obtained. (This section is like the bibliography in a published book).

Ref. No.	Document Title	Date of Release/ Publication	Document Source
PGBH01-2003-Proposal	Project Proposal	Oct 20, 2003	
IEEE-Std-830-1998	IEEE Recommended Practice for SRS	1998	IEEE Computer Society Standards

Ref. No.	Document Title	Date of Release/ Publication	Document Source

8. Appendice

8.1 Data Dictionary / Database Tables Schema Structure

The backend MySQL database architecture maintains strict row mappings based on the following specific schema designs:

Field Name	Data Type	Key Type	Null/Default	Description
book_id	VARCHAR(20)	Primary Key	NOT NULL	Unique non-repeating identifier for every physical book.
title	VARCHAR(250)	None	NOT NULL	The primary title of the book resource.
author_names	TEXT	None	NOT NULL	Supports single or multiple author names.
isbn_number	VARCHAR(30)	None	NOT NULL	International Standard Book Number of the book.
publisher_name	VARCHAR(150)	None	NULL Allowed	Publisher information of the book.
version_index	INT	None	DEFAULT 1	Edition/version number of the book.
serial_number	VARCHAR(50)	None	NOT NULL	Unique serial number assigned to the book copy.
available_copies	INT	None	DEFAULT 0	Number of copies currently available in inventory.

Field Name	Data Type	Key Type	Null/Default	Description
user_id	VARCHAR(20)	Primary Key	NOT NULL	Unique identifier for students, teachers, and administrators.
full_name	VARCHAR(150)	None	NOT NULL	Complete name of the user.
fathers_name	VARCHAR(150)	None	NULL Allowed	Father's name for identification purposes.
date_of_birth	DATE	None	NOT NULL	User's date of birth.
profile_role	VARCHAR(20)	None	NOT NULL	Defines whether the user is Student, Teacher, or Admin.
email_address	VARCHAR(100)	Unique	NOT NULL	User's email address.
class_session	VARCHAR(50)	None	NULL Allowed	Academic class and session details (for students).

